

Requirements Table : Community Lost & Found Matching Network

PES University: Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #59 | Navya Suresh | PES1UG24CS904

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001 (given)	Functional	The system shall match newly logged 'Found Item' posts with active 'Lost Item' reports based on category, colour tag, and location radius.	High	Pass: a match probability is calculated and a notification is dispatched to the potential owner. Fail: items of incompatible categories are matched.	Core matching functionality — the primary value proposition of the platform.
FR-002	Functional	The system shall allow an Owner to submit a Lost Item report containing a photo, text description, category, colour tag, last known location, and date lost.	High	Pass: a report with all required fields is saved and appears in the active Lost Item list within 2 seconds. Fail: a report missing a required field is rejected with a field-level error.	Structured, complete lost-item data is required before any matching can run.
FR-003	Functional	The system shall allow a Finder to submit a Found Item post containing a photo, text description, category, colour tag, and current geo-location.	High	Pass: a post with a valid geo-tag is saved and appears in the active Found Item list. Fail: a post submitted without geo-location is rejected.	Geo-location is required input for the proximity-radius matching in FR-001.
FR-004	Functional	The system shall notify an Owner via app/push notification within 60 seconds of a candidate match being identified for their Lost Item report.	Medium	Pass: the notification log timestamp is ≤ 60 seconds after the match was calculated. Fail: the notification is delayed beyond 60 seconds or is never sent.	Timely alerts increase the chance of successful, low-friction recovery.
FR-005	Functional	The system shall allow a Community Admin to flag a match as suspicious and remove the associated Lost/Found posts from active listings.	Medium	Pass: flagged posts are removed from public listings within 5 minutes and the action is written to the admin audit log. Fail: a flagged post remains publicly visible after the removal action.	Moderation is needed to prevent fraudulent claims and abuse of the contact-reveal flow.
NFR-001 (given)	Nonfunctional (Security)	Claimants must correctly answer private security questions set by the Finder before item-handover contact details are revealed.	High	Pass: benchmarking/security tests confirm the verification gate holds under simulated peak load and contact details are never exposed pre-verification. Fail: contact details are revealed without a successful verification.	Prevents impersonation and protects both parties' personal contact information.
NFR-002	Nonfunctional (Performance)	The system shall compute geo-proximity match candidates for a newly submitted post within 5 seconds when up to 10,000 active listings exist in the database.	Medium	Pass: load-test logs show a 95th-percentile match-computation time ≤ 5 seconds at 10,000 active listings. Fail: computation exceeds 5 seconds at that load.	Matching must stay fast as the community listing volume grows, or the core feature feels broken.